ICESI UNIVERSITY / COMPUTING & DISCRETE STRUCTURES III

ResumeLens

From skill mentions to verified profiles.

RESEARCH QUESTION

Can formal models keep profile decisions traceable?

Explicit rules. Original evidence.

Valentina Martinez Franco

Juan Pino

Resume Original text → **Regex** Evidence spans → **Transducers** Canonical symbols → **DFA** Profile decisions → **textX** Validated DSL → **HTML** Candidate report

01 Preserve & normalize

Alex Sample

Technical Skills

JS React.js · NodeJS · Postgres · Git

`text[238:240] == "JS"`

Regex preserves wording and positions.

Finite-state transducer (FST)

M = (Q, Σ, Γ, δ, ω, q0, F)



JS → js → JAVASCRIPT

ε is empty output; j is consumed.

JS path; state names abbreviated.

4 FSTs / 26 symbols / 56 aliases

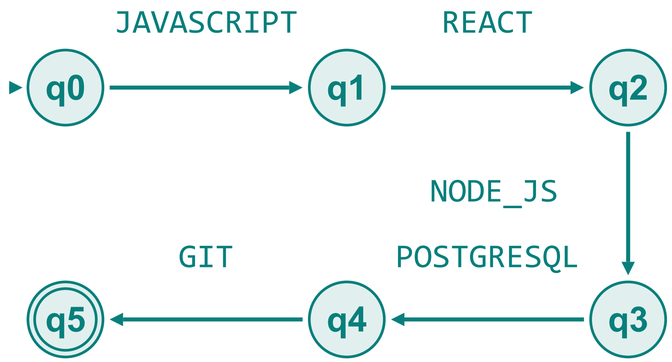
Filled check = exact case; cross = difference.

02 Recognize the profile

Deterministic finite automaton (DFA)

A = (Q, Σ, δ, q0, F)

Full Stack: two executed traces.



q5 accepts: all five groups are present.

Missing frontend: same q1, next NODE_JS



One token → one next state. No ε edges.

Other transitions: see complete models.

Full Stack / ML / Backend / Data Engineer

194 tests / 193 pass / 1 skip

03 Validate & evaluate

Extended Backus-Naur form (EBNF)

```
SkillEntry = "skill",
  SkillSymbol, ";" ;
```

textX syntax + profile consistency

Domain-specific language (DSL) excerpt

```
accepted_profiles {
  profile FULL_STACK_DEVELOPER;
}
```

Alex Sample HTML

✓ **Full Stack Developer accepted**

Separate synthetic evaluations

14/18

Exact cases



Preannotated dataset 18/18 valid pipelines

14/14 workflow regression (separate)

Traceable evidence. Deterministic decisions. Interpretation still has limits.

Known gaps: Rust, double negation, source attribution and Pandas wording. Synthetic results only.

Mohri (1997) Yu, Guan & Zhou (2005)

Code / formal models / evaluation methods